CMSC 27100-3 Due: Sunday, November 29, 2020

Problem Set 8

Instructor: Robert Rand

1 Probability Theory

Problem 1: Consider the event space of two die rolls. Find a group of events of A, B and C such that A, B and C are pairwise independent, but not independent. Show the joint probabilities.

Problem 2: On the morning of November 19th, 2020, Jason is 80% sure that he lives in Chicago, but thinks that there's a 10% chance each that he lives in New York or Los Angeles. He leaves his apartment at noon and it is over 15 degrees Celsius outside. The probability of it being over 15 degrees in each city is as follows:

- Chicago: 5%
- New York: 20%
- Los Angeles: 90%

Now how sure should Jason be that he lives in Chicago? What about New York and L.A.?

Problem 3: Connecting logic and probability theory, we could say that A implies B, i.e. $A \to B$, if $P(B \mid A) = 1$. Here we're going to define a weaker notion of A *hints at* B, written $A \rightsquigarrow B$, that holds when $(B \mid A) \geq P(B)$. (We will only concern ourselves with nonzero probability events.) Which of the following properties holds of the $\rightsquigarrow$ relation (prove them)?

1. Reflexivity
2. Symmetry
3. Transitivity
4. (Bonus) Contrapositivity: $(A \rightsquigarrow B) \to (\neg B \rightsquigarrow \neg A)$

2 Random Variables and Expectation

Problem 4: In the card game Bridge, the high card points system assigns the following point values to high cards:

- Ace: 4 points
- King: 3 points
- Queen: 2 points
- Jack: 1 point

All other cards get zero points.

1. Write a random variable X corresponding to this points assignment, where the probability space is a uniform distribution over a standard 52 card deck.
2. What is the probability that X is greater than one: $P(X > 1)$?
3. What is the expected value of a single card drawn from a deck of 52 cards?
4. What is the expected value of a hand of 13 cards (justify your answer)?